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ITAI 1378 – Computer Vision

Lab 06 Reflection Questions

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Reflection Questions

Part 1: Concepts

- 1. Explain the difference between classification, detection, and segmentation in your own words. Give one workforce example of each.**

Classification predicts what is in an image, detection identifies what objects are present and where they are located, and segmentation identifies the exact pixels that belong to each object. An example of classification is determining whether an X-ray shows pneumonia or not. An example of detection is identifying cars and pedestrians in a self-driving vehicle system. An example of segmentation is outlining a tumor in a medical image so doctors can see its exact shape and size.

- 2. What is Non-Maximum Suppression, and why does YOLO use it?**

Non-Maximum Suppression (NMS) is a process that removes duplicate bounding boxes around the same object. YOLO uses NMS to keep only the highest-confidence box when several boxes overlap heavily and represent the same object. This makes the detection results cleaner and easier to interpret.

- 3. In your own words, what does IoU measure?**

IoU measures how much a predicted bounding box overlaps with the correct bounding box. A higher IoU means the model located the object more accurately. An IoU of 1.0 means a perfect match, while an IoU of 0 means there is no overlap at all.

Part 2: Hands-on experience

- 4. When you ran YOLO11 on your own image, did it detect everything you expected? Did it detect anything that was wrong? Why do you think it succeeded or failed?**

When I ran YOLO11 on my image, it successfully detected the two people that were present in the image. It also detected a tie worn by one of the people. The detections were accurate and I did not notice any major errors. I think the model succeeded

because the image was clear and contained common objects that YOLO was trained to recognize.

- 5. When you compared detection and segmentation side by side, name one situation where the box would be enough and one situation where you would really need the mask.**

A bounding box would be enough for applications such as counting cars in a parking lot or detecting people in a security system. A segmentation mask would be necessary in applications such as medical image analysis or video editing because the exact shape and boundaries of the object are important.

- 6. SAM 2 does not output category labels. Why is that an important difference from YOLO?**

Sam 2 does not output category labels, while YOLO provides both labels and locations. This is important because YOLO can tell us what an object is, such as a person or a car, while SAM 2 only creates a mask around the object. SAM 2 is useful when we need precise segmentation, but YOLO is useful when we need object identification.

Part 3: Trade-offs

- 7. Confidence threshold trade-off. Give one application where you would want a low threshold (more sensitive) and one application where you would want a high threshold. Explain why.**

A low confidence threshold would be useful for a workplace safety alert system because it is important to detect as many people as possible, even if there are some false alarms. A high confidence threshold would be useful for a photo tagging application because it is better to avoid incorrect labels and only keep highly confident detections.

- 8. Precision versus recall. For a workplace safety system that detects unsafe behavior on a factory floor, do you want higher precision or higher recall? What about for a system that tags products in a customer's photo upload?**

For a workplace safety system, I would prefer higher recall because missing a person in a dangerous area could have serious consequences. For a product-tagging system, I would prefer higher precision because incorrect product labels could confuse customers and reduce trust in the system.

9. When would you reach for a specialist model like YOLO11-seg, and when would you reach for a foundation model like SAM 2?

I would use a specialist model such as YOLO11-seg when I need fast detection and segmentation of known object categories. I would use a foundation model such as SAM 2 when I need to segment unusual or unknown objects that may not belong to predefined categories. SAM 2 is more flexible, while YOLO11-seg is more specialized and efficient.

Part 4: Midterm project

10. What project area did you pick for your midterm? In one paragraph, describe: (a) what the application does, (b) where the data will come from, and (c) how you might extend it for the capstone.

For my midterm project, I chose a Fruit Freshness Detector. The application will use computer vision techniques to classify fruit images into six categories: fresh apple, fresh banana, fresh orange, rotten apple, rotten banana, and rotten orange. The goal is to help reduce food waste by automatically identifying whether fruit is fresh or spoiled. The dataset will come from publicly available fruit image datasets, and I plan to train and compare a custom CNN model with a transfer learning model such as ResNet-50. For the capstone version, I could expand the system to work with additional fruit types, process live video streams, and provide real-time freshness analysis for grocery stores or food distributors.